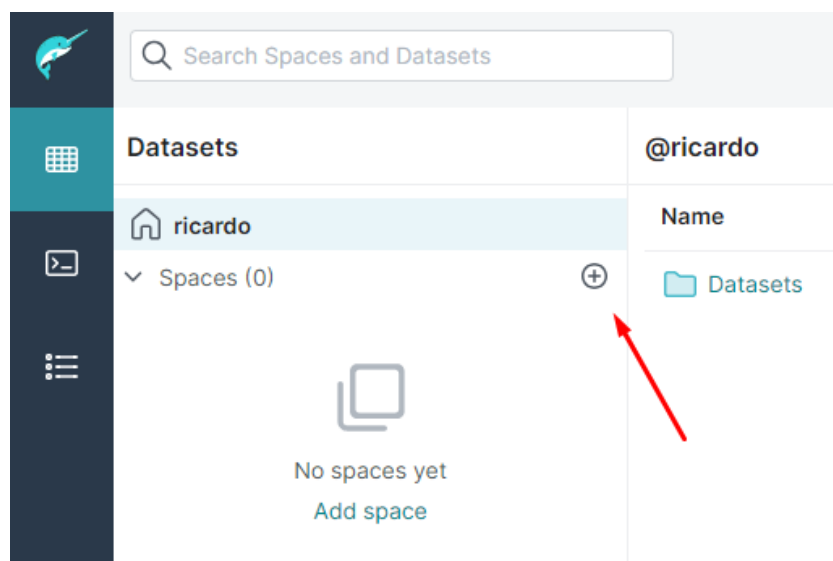


5075. Big Data aplicado - 1ª Evaluación (RA 1 – CE c)

Unidad Didáctica 1. Estrategias de ingestión y almacenamiento de datos en Big Data

Práctica 10. Transformando datos en Dremio.

Paso 1. Creamos un espacio que se llame “Ventas”.



Add space

Name

Ventas

Datasets	Ventas
 ricardo	
✓ Spaces (1) 	
 Ventas 0	

Paso 2. Realizamos una consulta a la tabla “Supermarket” y guardamos el dataset resultante dentro del espacio como un conjunto de datos virtuales.

Run

Preview

Hide SQL pane

Save as View

Context: (None selected)

1 SELECT "Invoice ID", "Product line", "Unit price", Quantity, Total FROM "@ricardo".Datasets.Supermarket

Query1

Add Column

Group By

Join

Filter Columns

5 Columns

Job: Run

Rows: 1,000

<1s

abc Invoice ID	abc Product line	abc Unit price	abc Quantity	abc Total
758-67-8428	Health and beauty	74.69	7	548.9715
226-31-3081	Electronic accessories	15.28	5	80.22
631-41-3108	Home and lifestyle	46.33	7	340.5255
123-19-1176	Health and beauty	58.22	8	489.048
373-73-7910	Sports and travel	86.31	7	634.3785
699-14-3026	Electronic accessories	85.39	7	627.6165

Hide SQL pane

Save as View

Context: (None selected)

Save Script as...

Save as View...

Save View As

Name

Ventas

Location

> @ricardo

▼ Ventas

Datasets	Ventas
ricardo	Name
▼ Spaces (1)	Ventas
Ventas 1	
Sources	

Paso 3. Creamos una wiki del dataset. Esto nos va a permitir tener la documentación centralizada, aportando contexto a los datos y trazabilidad, favoreciendo la colaboración y la comunicación.

107.142
246.4875
453.495
749.49
590.436
506.6355
457.443
172.2105
84.63
451.71

abc Total

▼ Wiki



No wiki content yet

[Edit wiki](#)

Wiki ×

Descripción General:

Este dataset contiene información relacionada con transacciones de facturación. Cada registro representa una transacción con detalles sobre la línea de productos, precios unitarios, cantidad y el total de la factura. Los campos principales y sus descripciones son los siguientes:

Campos:

****Invoice ID:****
Tipo de dato: Texto/Numérico.
Descripción: Identificador único de cada factura. Este campo asegura que cada transacción de facturación pueda ser rastreada individualmente.
Ejemplo: 750-67-8428.

****Product Line:****
Tipo de dato: Texto.
Descripción: Categoría o línea de productos asociados a la transacción. Indica el tipo de producto o servicio que se incluye en la factura.
Ejemplo: "Health and beauty".

****Unit Price:****
Tipo de dato: Numérico (Decimal).
Descripción: Precio unitario de un producto o servicio en la transacción, antes de aplicar cualquier descuento o impuesto.
Ejemplo: 19.99 (indica que cada unidad cuesta 19.99 de la moneda correspondiente).

****Quantity:****
Tipo de dato: Numérico (Entero).
Descripción: Cantidad de unidades del producto o servicio facturado. Este campo multiplica el precio unitario para calcular el subtotal.
Ejemplo: 3 (tres unidades).

Cancel Save

Rows: 1,000 <1s

abc Total
548.9715
80.22
340.5255
489.048
634.3785
627.6165
433.692
772.38
76.146
172.746
60.816
107.142
246.4875
453.495
749.49
590.436
506.6355
457.443
172.2105
84.63

Wiki

Descripción General:

Este dataset contiene información relacionada con transacciones de facturación. Cada registro representa una transacción con detalles sobre la línea de productos, precios unitarios, cantidad y el total de la factura. Los campos principales y sus descripciones son los siguientes:

Campos:

Invoice ID:
Tipo de dato: Texto/Numérico.
Descripción: Identificador único de cada factura. Este campo asegura que cada transacción de facturación pueda ser rastreada individualmente.
Ejemplo: 750-67-8428.

Product Line:
Tipo de dato: Texto.
Descripción: Categoría o línea de productos asociados a la transacción. Indica el tipo de producto o servicio que se incluye en la factura.
Ejemplo: "Health and beauty".

Paso 4. Revisamos los tipos de datos.

abc Unit price	...
74.69	Sort Ascending
15.28	Sort Descending
46.33	Sort Multiple...
58.22	Rename...
86.81	Drop
85.35	Convert Data Type...
68.84	Group By...
73.56	
36.26	
54.84	

Change Data Type

Convert to

Select Type

Select Type

010 Binary

Integer

Float

Date

abc Product line	abc Unit price
Health and beauty	74.69
Electronic accessories	15.28
Home and lifestyle	46.33
Health and beauty	58.22
Sports and travel	86.31

Change Data Type

Convert to

Float

Action for Non-matching Values

☒ Replace values with null

☐ Replace values with:

☐ Delete Rows

New Field Name

Unit price

Options

☒ Drop Source Field (Unit price)

Apply Preview Cancel Result based on sample dataset

abc Invoice ID	abc Product line	abc Unit price	## Unit price (new)
750-67-8428	Health and beauty	74.69	74.69
226-31-3081	Electronic accessories	15.28	15.28
631-41-3108	Home and lifestyle	46.33	46.33
123-19-1176	Health and beauty	58.22	58.22
373-73-7910	Sports and travel	86.31	86.31
699-14-3026	Electronic accessories	85.39	85.39
355-53-5943	Electronic accessories	68.84	68.84
315-22-5665	Home and lifestyle	73.56	73.56
665-32-9167	Health and beauty	36.26	36.26

## Unit price	...	# Quantity	...	## Total	...
74.69		7		548.9715	
15.28		5		80.22	
46.33		7		340.5255	
58.22		8		489.048	
86.31		7		634.3785	
85.39		7		627.6165	

```
SELECT "Invoice ID", "Product line", CONVERT_TO_FLOAT("Unit price", 1, 1, 0) AS "Unit price", CONVERT_TO_INTEGER(Quantity, 1, 1, 0) AS Quantity,
CONVERT_TO_FLOAT(Total, 1, 1, 0) AS Total
FROM Ventas.Ventas AS Ventas;
```

Paso 5. Limpiamos los espacios de las cadenas y capitalizamos el texto.

abc Product line	...	## Unit price
Health	Sort Ascending	
Electro	Sort Descending	
Home ar	Sort Multiple...	
Health	Rename...	
Sports	Drop	
Electro	Convert Data Type...	
Electro	Group By...	
Home ar	Convert Case...	
Health	Trim Whitespace...	
Food ar	Calculated Field...	
Fashio		
Electro		
Electro		
Food ar		

Convert Case

Options

- ☐ UPPERCASE
☐ lowercase
☒ Title Case

New Field Name

Product line

Options

- ☒ Drop Source Field (Product line)

Apply

Preview

Cancel

⚠ Result based on sample dataset

abc Invoice ID	abc Product line	abc Product line (new)
750-67-8428	Health and beauty	Health And Beauty
226-31-3081	Electronic accessories	Electronic Accessories
631-41-3108	Home and lifestyle	Home And Lifestyle
123-19-1176	Health and beauty	Health And Beauty
373-73-7910	Sports and travel	Sports And Travel
699-14-3026	Electronic accessories	Electronic Accessories
355-53-5943	Electronic accessories	Electronic Accessories
315-22-5665	Home and lifestyle	Home And Lifestyle
665-32-9167	Health and beauty	Health And Beauty

Trim Whitespace

Options

- ☒ Trim from both sides
☐ Trim from the start (Trim Left)
☐ Trim from the end (Trim Right)

New Field Name

Product line

Options

- ☒ Drop Source Field (Product line)

Apply Preview Cancel  Result based on sample dataset

abc Invoice ID	abc Product line	abc Product line (new)
750-67-8428	Health And Beauty	Health And Beauty
226-31-3081	Electronic Accessories	Electronic Accessories
631-41-3108	Home And Lifestyle	Home And Lifestyle
123-19-1176	Health And Beauty	Health And Beauty
373-73-7910	Sports And Travel	Sports And Travel
699-14-3026	Electronic Accessories	Electronic Accessories
355-53-5943	Electronic Accessories	Electronic Accessories
315-22-5665	Home And Lifestyle	Home And Lifestyle

Paso 6. Vamos a separar por partes el Id de las facturas y luego lo vamos a volver a unir.

abc Invoice ID	...
750-67-	Sort Ascending
226-31-	Sort Descending
631-41-	Sort Multiple...
123-19-	Rename...
373-73-	Drop
699-14-	Convert Data Type...
355-53-	Group By...
315-22-	Convert Case...
665-32-	Trim Whitespace...
692-92-	Calculated Field...
351-62-	Extract Text...
529-56-	Replace Text...
365-64-	Split...
252-56-	
829-34-	
299-46-	
656-95-	
765-26-	

Replace Extract **Split** Keep Only Exclude

Edit Selection

Fixed String ▾

-

☐ Ignore Case



■ 1000 matched values ■ 0 unmatched values

Position

All ▾

Max Fields

10

New Field Name

Invoice ID_1

Apply

Preview

Cancel

⚠ Result based on sample data

abc Invoice ID	{ } Invoice ID_1
750-67-8428	["750","67","8428"]
226-31-3081	["226","31","3081"]
631-41-3108	["631","41","3108"]

...	{ } Invoice ID_1	...
	[" Rename...	
	[" Drop	
	[" Convert Data Type...	
	[" Unnest	
	[" Extract Element(s)...	
	[" Calculated Field...	
	["665", "32", "9167"]	
	["692", "92", "5582"]	
	["351", "62", "0822"]	

Replace

Extract

Split

Keep Only

Exclude

Extract

☒ Single

☐ Multiple

Element

0

1000 matched values

0 unmatched values

New Field Name

Invoice ID_1_1

Options

☐ Drop Source Field (Invoice ID_1)

Apply

Preview

Cancel

⚠

Result based on sample dataset

abc Invoice ID	{.} Invoice ID_1	abc Invoice ID_1_1
750-67-8428	["750","67","8428"]	750
226-31-3081	["226","31","3081"]	226
631-41-3108	["631","41","3108"]	631
123-19-1176	["123","19","1176"]	123

...	{.} Invoice ID_1	...
	["	Rename...
	["	Drop
	["	Convert Data Type...
	["	Unnest
	["	Extract Element(s)...
	["	Calculated Field...
	["665", "32", "9167"]	
	["692", "92", "5582"]	
	["351", "62", "0822"]	

Add Calculated Field

```
1 CONCAT("Invoice ID_1"[0], '_', "Invoice ID_1"[1], '_', "Invoice ID_1"[2])
```

New Field Name

Options

Invoice ID_1_1

☐ Drop Source Field (Invoice ID_1)

Apply

Preview

Cancel

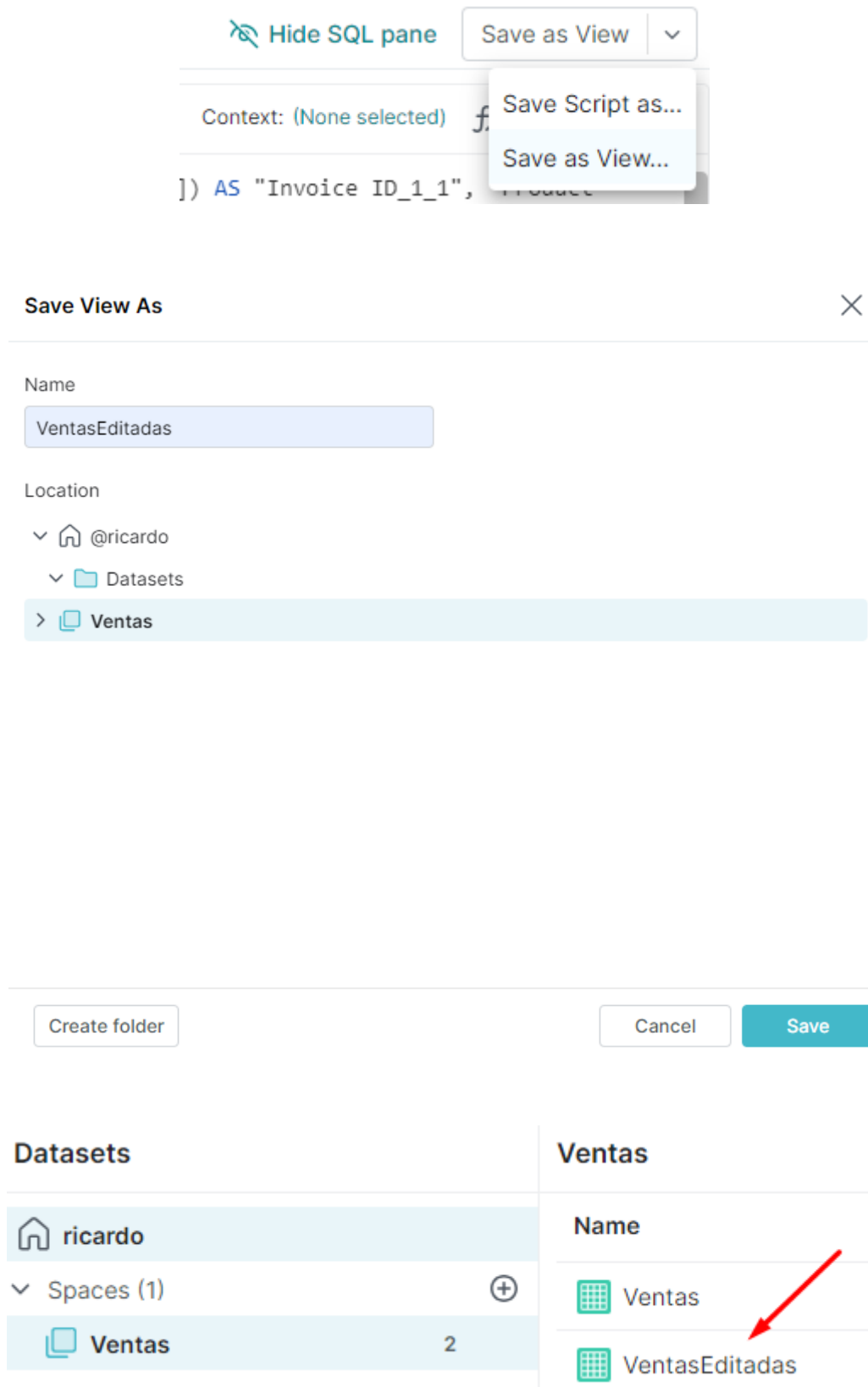
 Result based on sample dataset

abc Invoice ID	{.} Invoice ID_1	abc Invoice ID_1_1
750-67-8428	["750", "67", "8428"]	750_67_8428
226-31-3081	["226", "31", "3081"]	226_31_3081
631-41-3108	["631", "41", "3108"]	631_41_3108

```
SELECT "Invoice ID", "Invoice ID_1", CONCAT("Invoice ID_1"[0], '_', "Invoice ID_1"[1], '_', "Invoice ID_1"[2]) AS "Invoice ID_1_1", "Product line", "Unit price", Quantity, Total
FROM (
  SELECT "Invoice ID", regexp_split("Invoice ID", '\Q-\E', 'ALL', 10) AS "Invoice ID_1", trim(both ' ' from TITLE("Product line")) AS "Product line", CONVERT_TO_FLOAT("Unit price", 1, 1, 0) AS "Unit price", CONVERT_TO_INTEGER(Quantity, 1, 1, 0) AS Quantity, CONVERT_TO_FLOAT(Total, 1, 1, 0) AS Total
  FROM Ventas.Ventas AS Ventas
) nested_0;
```

abc Invoice ID	{.} Invoice ID_1	abc Invoice ID_1_1	abc Product line	## Unit price	# Quantity
750-67-8428	["750", "67", "8428"]	750_67_8428	Health And Beauty	74.69	
226-31-3081	["226", "31", "3081"]	226_31_3081	Electronic Accessories	15.28	
631-41-3108	["631", "41", "3108"]	631_41_3108	Home And Lifestyle	46.33	
123-19-1176	["123", "19", "1176"]	123_19_1176	Health And Beauty	58.22	
373-73-7910	["373", "73", "7910"]	373_73_7910	Sports And Travel	86.31	
699-14-3026	["699", "14", "3026"]	699_14_3026	Electronic Accessories	85.39	
355-53-5943	["355", "53", "5943"]	355_53_5943	Electronic Accessories	68.84	
315-22-5665	["315", "22", "5665"]	315_22_5665	Home And Lifestyle	73.56	
665-32-9167	["665", "32", "9167"]	665_32_9167	Health And Beauty	36.26	
692-92-5582	["692", "92", "5582"]	692_92_5582	Food And Beverages	54.84	
351-62-0822	["351", "62", "0822"]	351_62_0822	Fashion Accessories	14.48	
529-56-3974	["529", "56", "3974"]	529_56_3974	Electronic Accessories	25.51	
365-64-0515	["365", "64", "0515"]	365_64_0515	Electronic Accessories	46.95	
252-56-2699	["252", "56", "2699"]	252_56_2699	Food And Beverages	43.19	
829-34-3910	["829", "34", "3910"]	829_34_3910	Health And Beauty	71.38	
299-46-1805	["299", "46", "1805"]	299_46_1805	Sports And Travel	93.72	
656-95-9349	["656", "95", "9349"]	656_95_9349	Health And Beauty	68.93	
765-26-6951	["765", "26", "6951"]	765_26_6951	Sports And Travel	72.61	

Paso 7. Guardamos los cambios realizados.



Save View As ×

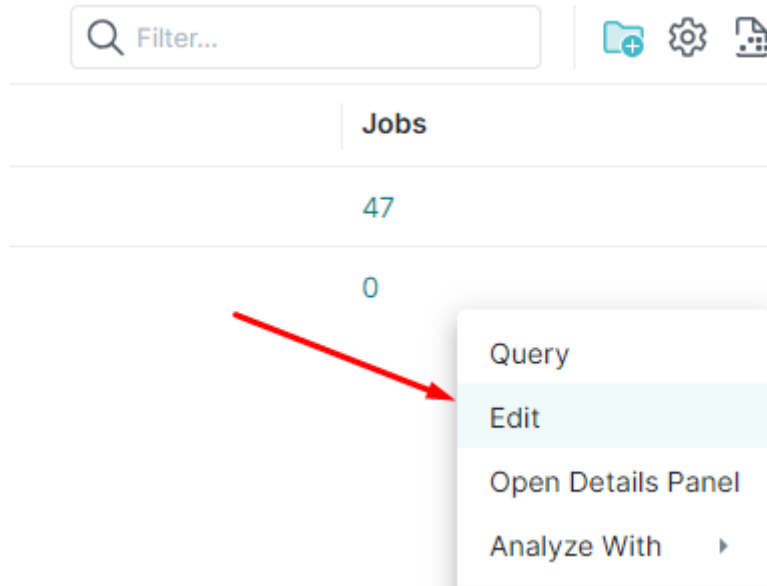
Name
VentasEditadas

Location
▼ @ricardo
▼ Datasets
> Ventas

Create folder Cancel Save

Datasets	Ventas
ricardo	Name
▼ Spaces (1) ⊕	Ventas
Ventas 2	VentasEditadas

Si quisiéramos editar el dataset virtual podremos hacerlo de la siguiente manera:



Run Preview Hide SQL pane Save View

Context: (None selected)

```

1 SELECT "Invoice ID", "Invoice ID_1", CONCAT("Invoice ID_1[0]", '-', "Invoice ID_1[1]", '-', "Invoice ID_1[2]") AS "Invoice ID_1_1", "Product line", "Unit price", Quantity, Total
2 FROM (
3   SELECT "Invoice ID", regexp_split("Invoice ID", '[Q-VE]', 'ALL', 10) AS "Invoice ID_1", trim(both ' ' from TITLE("Product line")) AS "Product line", CONVERT_TO_FLOAT("Unit price", 1, 1, 0) AS "Unit price",
4   CONVERT_TO_INTEGER(Quantity, 1, 1, 0) AS Quantity, CONVERT_TO_FLOAT(Total, 1, 1, 0) AS Total
5   FROM Ventas.Ventas AS Ventas
6   ) nested_0;

```

Query1

Add Column Group By Join Filter Columns 7 Columns Job: Run Rows: 1,000 <1s

Invoice ID	Invoice ID_1	Invoice ID_1_1	Product line	Unit price	Quantity	Total
758-67-8428	["758", "67", "8428"]	758_67_8428	Health And Beauty	74.69	7	546.9715
126-31-3881	["126", "31", "3881"]	126_31_3881	Electronic Accessories	15.28	5	88.22
631-41-3188	["631", "41", "3188"]	631_41_3188	Home And Lifestyle	46.33	7	348.5255
123-19-1176	["123", "19", "1176"]	123_19_1176	Health And Beauty	58.22	8	489.848
373-73-7918	["373", "73", "7918"]	373_73_7918	Sports And Travel	86.31	7	624.3785
699-14-3826	["699", "14", "3826"]	699_14_3826	Electronic Accessories	85.39	7	627.4165
355-53-5943	["355", "53", "5943"]	355_53_5943	Electronic Accessories	68.84	6	433.692
315-22-5665	["315", "22", "5665"]	315_22_5665	Home And Lifestyle	73.56	10	772.38
665-32-9187	["665", "32", "9187"]	665_32_9187	Health And Beauty	36.26	2	76.146
692-92-5582	["692", "92", "5582"]	692_92_5582	Food And Beverages	54.84	3	172.746
351-62-8822	["351", "62", "8822"]	351_62_8822	Fashion Accessories	14.48	4	68.816
529-56-3974	["529", "56", "3974"]	529_56_3974	Electronic Accessories	25.51	4	187.142